Glass Ceiling Unshattered

Asymmetric Gender Culture among Immigrants in Different German Tracks

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One slide

- Immigrant families carry an aspiration premium: higher ambitions than comparable natives, net of SES.
- In Germany, that premium helps immigrant *boys* outperform in **Gymnasium**.
- But it becomes a liability for immigrant girls in the same track.
- **Comprehensive schools protect them. Gymnasium compounds the gap.**
- Mechanism: a *gendered aspiration threshold* activated only under high institutional demand.

Motivation

The puzzle

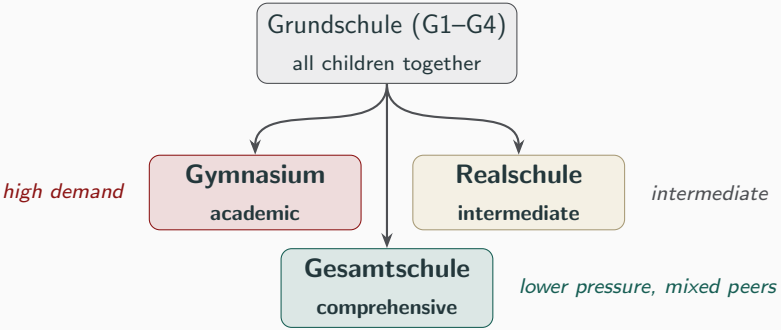
Immigrant aspiration paradox (Kao & Tienda 1995; Feliciano & Lanuza 2017):

- Immigrant families hold higher educational aspirations than natives, conditional on SES.
- Immigrant boys who make it to the academic track outperform native boys.

But this is not gender-neutral.

When does the immigrant aspiration premium become a liability for girls?

The German three-track system



Students sort after G4 on the basis of teacher recommendations and parental choice. Once sorted, they face radically different peer environments, teacher expectations, and curricular demands.

Hypothesis

The glass ceiling hypothesis

Two observations the literature supplies separately:

1. **Immigrant aspiration paradox** — immigrant families hold higher ambitions than natives.
2. **Traditional gender norms** — carried across generations, with asymmetric enforcement on daughters.

Prediction: a *gendered aspiration threshold* that activates only when the institutional environment demands sustained, visible academic commitment.

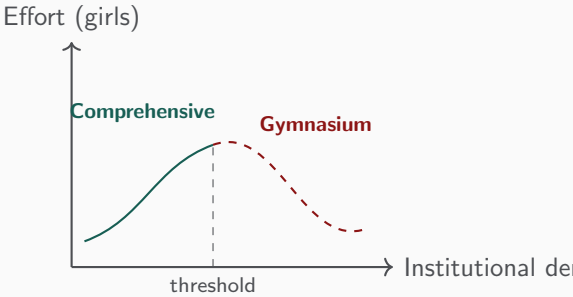
- **Gymnasium:** high demand $\Rightarrow$ ceiling binds $\Rightarrow$ immigrant girls fall behind.
- **Comprehensive:** lower demand $\Rightarrow$ ceiling does not bind $\Rightarrow$ immigrant girls protected.

The mechanism, intuitively

Not a story about lack of support.

It is a story about support being conditional on institutional demand.

- When demand is moderate, immigrant culture is an asset for both genders.
- When demand is high (Gymnasium), it becomes an asset for boys and a liability for girls.



Schematic: effort collapses once demand crosses the family's gendered ceiling.

Data & Design

Data: pooled NEPS starting cohorts 2 and 3

- **SC2** — tracked from G4 into secondary; G7 and G9 math assessments. $N = 1{,}311$.
- **SC3** — starts G5, followed through G7, G9. $N = 2{,}193$.
- Common IRT anchoring $\Rightarrow$ pooled $N = 3{,}504$.

	N	% Female	% Immigrant	% High SES	Mean G7	Mean G9
Comprehensive	424	54.0	22.9	62.0	−0.20	−0.56
Realschule	780	45.1	17.3	56.9	+0.33	−0.36
Gymnasium	2,300	52.0	20.3	79.5	+1.09	+0.59

Identification: value-added = DiD on the gender gap

Within each track j:

$$\text{mag9}_{ij} = \alpha_j + \beta_{1j}\,\text{mag7}_{ij} + \beta_{2j}\,\text{Female}_i + \gamma_j X_i + \varepsilon_{ij}$$

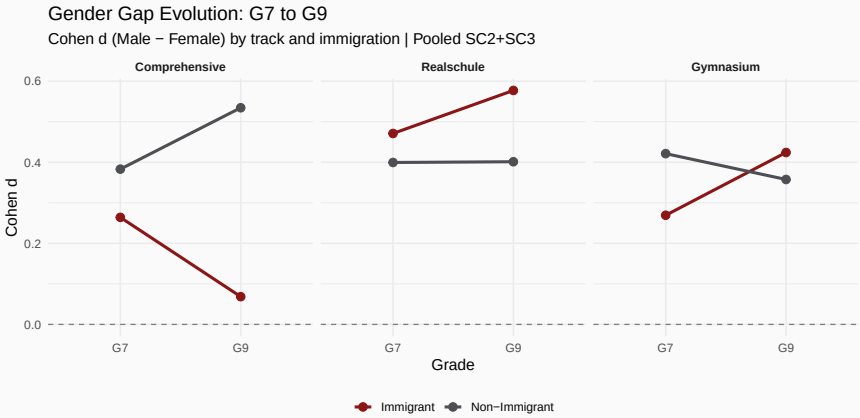
Consider a boy and a girl who score identically at G7.

- If $\beta_2 < 0$, the girl scores lower at G9 *despite starting at the same level*.
- First difference: across gender. Second difference: across time (absorbed by G7).
- β_2 = the gender gap that *develops* during secondary school, net of prior achievement.

Four nested specifications: Baseline $\rightarrow$ + demographics $\rightarrow$ + Female$\times$Immigrant $\rightarrow$ + Female$\times$SES.

Main results

The gap widens in every track — steepest for immigrants in Gymnasium



Cohen's *d* (Male − Female). Positive = male advantage. Gap widens from G7 to G9 in every cell; immigrant Gymnasium shows the steepest divergence.

Headline coefficients (value-added, Female coefficient)

Female at G9 conditional on G7:

- **Comprehensive:** -0.211^{*} (Female $\times$ Immigrant $= +0.402^{*}$) $\Rightarrow$ immigrant girls protected.
- **Realschule:** -0.177^{**} (Female $\times$ Immigrant $= -0.112$, n.s.) $\Rightarrow$ uniform gap.
- **Gymnasium:** -0.158^{***} (Female $\times$ Immigrant $= -\mathbf{0.184}^{*}$) $\Rightarrow$ immigrant girls compounded.

Total female effect for immigrant girls in Gymnasium: $-0.121 + (-0.184) = -\mathbf{0.305}$.
Roughly double the gap for non-immigrant girls.

Female $\times$ High SES is insignificant in all three tracks.
The mechanism is cultural, not material.

Matched sample: SC2's pre-tracking leverage

Matching on G4 (pre-tracking) math ability

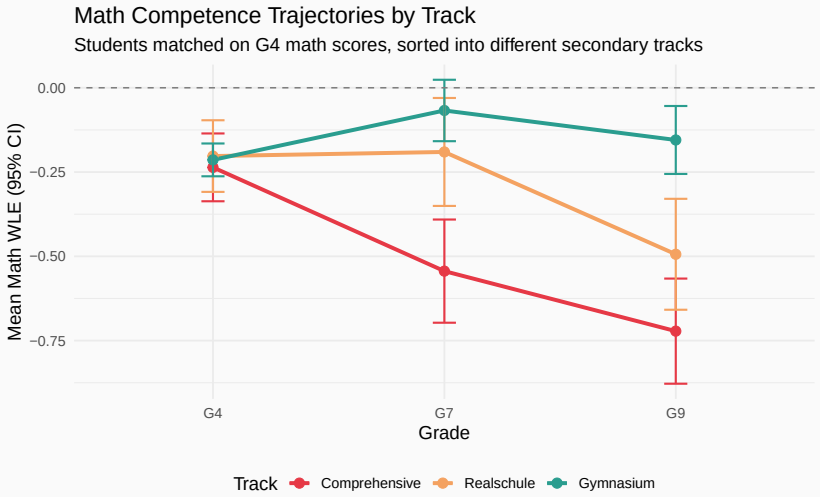
Concern: value-added conditions on G7, but G7 is already post-sorting.

Fix: use SC2's G4 math scores — the last measurement before any track decision.

- Match on G4 within Q10–Q90 common support; trim Gymnasium upper tail to equalize means.
- Matched $N = 626$. G4 balance: $F(2, 623) = 0.151$, $p = 0.86$.

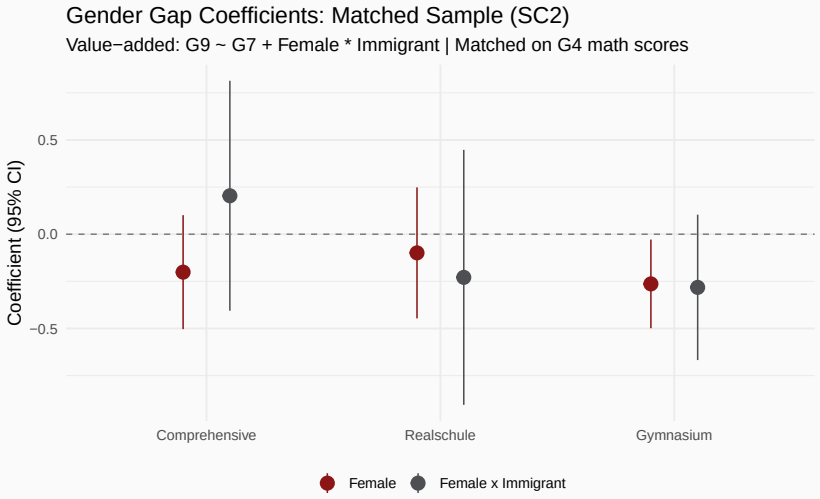
Students who scored identically at G4 can now be compared across tracks they subsequently entered.

Divergent trajectories from identical starting points



Matched on G4 math, students sorted into different tracks diverge by ~ 0.57 SD by G9.

The gap doubles in the matched Gymnasium



Matched Gymnasium Female coefficient: $-\mathbf{0.355}$ (more than double the pooled estimate).

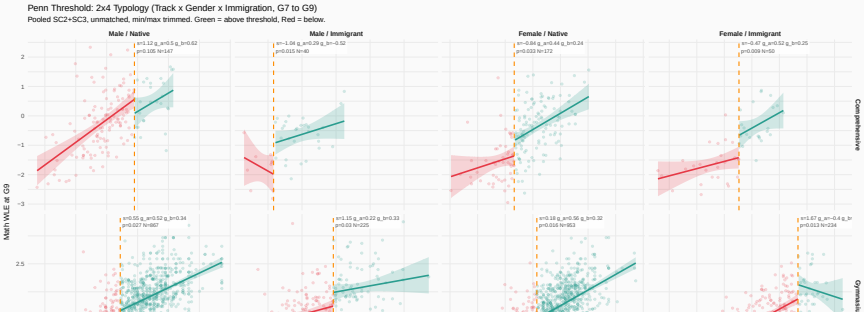
Structural estimation

Penn (2017) threshold: one cell breaks the pattern

$$s_{t+1}^i = \begin{cases} a + \gamma_1 s_t^i + \varepsilon_i & \text{if } s_t^i > \bar{s} \\ a + \gamma_2 s_t^i + \varepsilon_i & \text{if } s_t^i \leq \bar{s} \end{cases}$$

with Penn's prediction $\gamma_1 < \gamma_2$ (lock-in below, mobility above).

Estimated by Hansen (2000) grid search on $\mathrm{G7} \rightarrow \mathrm{G9}$.



The only cell with the aspiration trap

Seven of eight cells: $\gamma_{\text{above}} > \gamma_{\text{below}}$ or symmetric — effort compounds above, mobility below. Normal.

One cell inverts completely: Gymnasium Immigrant Girls.

- $\gamma_{\text{above}} = -0.402$, $\gamma_{\text{below}} = +0.668$, $F = 4.46$, $p = 0.013$.
- Below the threshold: scores lock in.
- Above the threshold: the few girls who clear it regress.

Effort above the cutoff is **punished** *rather than rewarded.*
This is the structural signature of the glass ceiling.

Interpretation

Why the Dustmann et al. (2016) null isn't the final word

- Dustmann et al.: cross-state variation in tracking age $\Rightarrow$ zero average long-term wage effect.
- But the aggregate null pools across demographic subgroups.
- Tracking simultaneously elevates immigrant boys and depresses immigrant girls.
- The average washes out; the distribution does not.

The glass ceiling is a compositional heterogeneity story the aggregate obscures.

Why not teacher discrimination or SES

- **Not peer rejection / "acting white."** The mechanism is family-driven, not peer-driven (Dahl et al. 2022).
- **Not SES.** Female $\times$ High SES is null in all three tracks — books don't buffer girls.
- **Not track mis-assignment alone.** Matched-sample evidence survives pre-tracking balance.
- **What's left:** the interaction of traditional gender norms with high institutional demand.

Contribution

Connects four literatures through one mechanism:

- **Aspirations** (Ray; Genicot & Ray) — gendered threshold.
- **Identity** (Akerlof & Kranton; Shayo) — gender-differentiated prescriptions.
- **Immigrant paradox** (Feliciano & Lanuza) — asymmetric premium.
- **Tracking** (Hanushek & Woessmann; Van de Werfhorst & Mijs) — institutional amplifier.

The glass ceiling is not discrimination, not poverty, not peer rejection.
It is cultural expectation triggered by institutional pressure.

Policy implications

- **Comprehensive schooling protects immigrant girls** from the aspiration trap — not because Gymnasium is harmful per se, but because the intersection with traditional norms is.
- Mixed-ability classrooms within Gymnasium, mentoring targeting immigrant girls, or norm-setting interventions may deactivate the ceiling *without* dismantling the track.
- Material-side policies (scholarships, textbooks) will not close this gap. The binding constraint is cultural permission to aspire, not resources.

Thank you.

Questions?

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